Local-First Privacy AI

This document is a test file for the Local-First Privacy AI Backend. It contains several paragraphs of plain text so that the PDF loader can extract meaningful content and the text splitter can produce more than one chunk. All processing in this project happens on the local machine, which means documents never leave the user's computer.

A local-first AI system runs its language model directly on the same device that stores the data. There are no external API calls and no cloud storage. This design greatly reduces the risk of sensitive information leaking to third parties, and it allows the application to work without an internet connection after the models have been downloaded.

Retrieval augmented generation, often shortened to RAG, is the technique used by this backend. First, a document is loaded and divided into small overlapping pieces called chunks. Each chunk is converted into a numeric vector by an embedding model. The vectors are stored in a local vector database. When a user asks a question, the question is also converted into a vector, and the database returns the chunks whose vectors are most similar.

The retrieved chunks are then inserted into the prompt that is sent to the language model. This gives the model relevant context from the user's own documents, which makes the answer more accurate and reduces invented details. Because every component runs locally, the full pipeline remains private.

The upload endpoint accepts a PDF file, stores it in a temporary location, reads its pages with the PyPDFLoader helper, and splits the extracted text with the recursive character text splitter. The temporary file is deleted immediately after processing. The endpoint reports how many pages and chunks were produced so that the ingestion step can be verified before the data is embedded and stored.

This final paragraph exists simply to add enough text to the test document. It repeats that privacy, local execution, simple modular Python code, and fast local inference are the main goals of the project. The backend should remain easy for a beginner to read and extend.